

Nonhomogeneous Equations & Undetermined Coefficients

Ordinary Differential Equations — Cheat Sheet & Review Guide

Topic 1: Trial Forms, Superposition, and Resonance

PART I Condensed Cheat Sheet

1 The Nonhomogeneous Linear Equation

We study constant-coefficient linear equations of the form

The Standard Form

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x), \quad g(x) \neq 0.$$

The associated *homogeneous* equation replaces $g(x)$ with 0.

The term $g(x)$ is the **forcing function** (or input/driving term). The whole subject splits into two pieces: a solution of the homogeneous problem and one extra solution that “absorbs” the forcing.

Theorem 1 (Structure of the General Solution). *If y_p is any particular solution of the nonhomogeneous equation and $y_c = c_1 y_1 + \cdots + c_n y_n$ is the general solution of the associated homogeneous equation, then the general solution of the nonhomogeneous equation is*

General Solution

$$y(x) = y_c(x) + y_p(x) = \underbrace{c_1 y_1 + \cdots + c_n y_n}_{\text{complementary}} + \underbrace{y_p}_{\text{particular}}.$$

Workflow. (1) Solve the characteristic equation to obtain y_c . (2) Build a trial y_p from $g(x)$. (3) Adjust for overlap with y_c . (4) Substitute to pin down the coefficients. (5) Add: $y = y_c + y_p$. (6) Apply initial conditions *last*, to the full y .

2 Method of Undetermined Coefficients

The method applies precisely when $g(x)$ is built from terms whose derivatives recycle into a finite family: **polynomials**, **exponentials** $e^{\alpha x}$, **sines/cosines** $\cos \beta x$, $\sin \beta x$, **and finite products of these**. It fails for inputs like $\tan x$, $\ln x$, or $1/x$ (use variation of parameters there).

The idea. Guess y_p with the same “shape” as $g(x)$ but with unknown coefficients, then force the equation to hold.

Master Table of Trial Forms

Forcing term $g(x)$	Trial particular solution y_p
$P_n(x) = b_n x^n + \cdots + b_0$	$x^s(A_n x^n + \cdots + A_1 x + A_0)$
$Ce^{\alpha x}$	$x^s A e^{\alpha x}$
$C \cos \beta x$ or $C \sin \beta x$	$x^s(A \cos \beta x + B \sin \beta x)$
$P_n(x)e^{\alpha x}$	$x^s e^{\alpha x}(A_n x^n + \cdots + A_0)$
$P_n(x)e^{\alpha x} \cos \beta x$ (or $\sin \beta x$)	$x^s e^{\alpha x}[(A_n x^n + \cdots + A_0) \cos \beta x + (B_n x^n + \cdots + B_0) \sin \beta x]$

Here s is the **multiplicity** of the associated root in the characteristic equation (see §4). If there is no overlap with y_c , take $s = 0$ (so $x^s = 1$).

Two non-negotiable habits.

- Always include the *complete* family. A bare $\cos \beta x$ input still requires *both* $\cos \beta x$ and $\sin \beta x$ in the guess, because differentiation mixes them.
- For a polynomial of degree n , include *every* power $x^n, \dots, x^1, x^0$, even powers absent from $g(x)$.

3 The Superposition Principle

Theorem 2 (Superposition for Particular Solutions). If $y_p^{(i)}$ solves $L[y] = g_i(x)$ for each i , then $y_p^{(1)} + y_p^{(2)} + \cdots + y_p^{(k)}$ solves $L[y] = g_1(x) + g_2(x) + \cdots + g_k(x)$.

In practice: when $g(x)$ is a *sum* of admissible pieces, split it, solve each piece separately, then add the particular solutions.

Superposition

$$L[y] = g_1 + g_2 + \cdots + g_k \implies y_p = y_p^{(1)} + y_p^{(2)} + \cdots + y_p^{(k)}.$$

Caution. Test each piece g_i *separately* for overlap with y_c . One piece may resonate while another does not.

4 Overlap & Resonance — The Modification Rule

The trial table fails when a guessed term is *already* a homogeneous solution: substituting it returns 0, never the forcing term. The cure is to multiply by the lowest power of x that clears the duplication.

Theorem 3 (Modification / Multiplicity Rule). If any term of the standard trial y_p duplicates a term of y_c , multiply that entire trial block by x^s , where s is the smallest positive integer that removes the duplication. Equivalently, s equals the multiplicity of the corresponding root of the characteristic equation.

Resonance Bookkeeping

- Forcing $e^{\alpha x}$: let $s =$ multiplicity of α as a real root. Trial $\rightarrow x^s A e^{\alpha x}$.
- Forcing $\cos \beta x / \sin \beta x$: let $s =$ multiplicity of $\pm i\beta$. Trial $\rightarrow x^s (A \cos \beta x + B \sin \beta x)$.
- Forcing $e^{\alpha x} \cos \beta x$: let $s =$ multiplicity of $\alpha \pm i\beta$.

Physical reading. When the driving frequency matches a natural frequency, $s \geq 1$ and the solution carries a factor of x (or t): the amplitude grows without bound. This is **pure resonance**.

5 Procedure Checklist

1. Find y_c from the characteristic equation; *record its roots and their multiplicities*.
2. Decompose $g(x)$ into admissible pieces (superposition).
3. Write each standard trial form (complete families!).
4. Compare with y_c ; multiply offending blocks by x^s .
5. Differentiate, substitute, and match coefficients of like terms.
6. Assemble $y = y_c + y_p$; apply initial conditions to y , never to y_c alone.

PART II

Review Guide — Practice Set

Instructions. Problems 1–4 ask only for the *correct trial form* (state y_c , check for overlap, and write y_p with undetermined coefficients — do *not* solve for them). Problems 5–10 ask for complete solutions.

- Q1. (Classification — polynomial input.)** For $y'' + 4y = 3x^2 - 1$, state y_c and write the correct trial form for y_p .
- Q2. (Classification — exponential, check resonance.)** For $y'' - y' - 2y = 5e^{2x}$, state y_c and write the correct trial form for y_p .
- Q3. (Classification — trigonometric input.)** For $y'' + 9y = \cos 2x$, state y_c and write the correct trial form for y_p .
- Q4. (Classification — trigonometric resonance.)** For $y'' + 9y = \sin 3x$, state y_c and write the correct trial form for y_p .
- Q5. (Solve — polynomial forcing.)** Find the general solution of $y'' - 3y' + 2y = 4x$.
- Q6. (Solve — exponential forcing.)** Find the general solution of $y'' - y = 3e^{2x}$.
- Q7. (Solve — superposition.)** Find the general solution of $y'' + y = 4x + 2e^x$.
- Q8. (Solve — exponential resonance, double root.)** Find the general solution of $y'' - 4y' + 4y = 6e^{2x}$.
- Q9. (Solve — trigonometric resonance.)** Find the general solution of $y'' + 4y = \sin 2x$.
- Q10. (Application — driven oscillator with resonance, IVP.)** An undamped mass–spring system is modeled by

$$y'' + 16y = 2 \cos 4t, \quad y(0) = 0, \quad y'(0) = 0.$$

Solve the initial value problem and describe the long-term behavior of the response.

PART III

Answer Key & Worked Solutions

Solution 1 — Polynomial input

The characteristic equation is $r^2 + 4 = 0$, so $r = \pm 2i$ and

$$y_c = c_1 \cos 2x + c_2 \sin 2x.$$

The forcing $3x^2 - 1$ is a degree-2 polynomial, so the standard guess is a *complete* quadratic. Since y_c contains no polynomial terms, there is no overlap and $s = 0$:

$$y_p = Ax^2 + Bx + C$$

(Note: include the Bx and C terms even though g has no x^1 term.)

Solution 2 — Exponential with resonance check

Characteristic equation: $r^2 - r - 2 = (r - 2)(r + 1) = 0$, giving $r = 2, -1$, so

$$y_c = c_1 e^{2x} + c_2 e^{-x}.$$

The forcing is $5e^{2x}$. The standard guess Ae^{2x} *duplicates* the y_c term $c_1 e^{2x}$. Since $r = 2$ is a root of multiplicity 1, take $s = 1$:

$$y_p = Axe^{2x}$$

Solution 3 — Trigonometric input

Characteristic equation: $r^2 + 9 = 0 \Rightarrow r = \pm 3i$, so

$$y_c = c_1 \cos 3x + c_2 \sin 3x.$$

The forcing $\cos 2x$ corresponds to $\pm 2i$, which is *not* a characteristic root ($\pm 3i \neq \pm 2i$). No overlap, $s = 0$, and we still need the full trig family:

$$y_p = A \cos 2x + B \sin 2x$$

Solution 4 — Trigonometric resonance

As in Q3, $r = \pm 3i$ and $y_c = c_1 \cos 3x + c_2 \sin 3x$. Now the forcing $\sin 3x$ corresponds to $\pm 3i$, which *is* a characteristic root (multiplicity 1). The standard guess $A \cos 3x + B \sin 3x$ duplicates y_c , so multiply by x^s with $s = 1$:

$$y_p = x(A \cos 3x + B \sin 3x)$$

Solution 5 — Polynomial forcing

Complementary solution. $r^2 - 3r + 2 = (r - 1)(r - 2) = 0 \Rightarrow r = 1, 2$, so

$$y_c = c_1 e^x + c_2 e^{2x}.$$

Trial. Forcing $4x$ is linear; no polynomial overlap, so

$$y_p = Ax + B, \quad y'_p = A, \quad y''_p = 0.$$

Substitute into $y'' - 3y' + 2y$:

$$\begin{aligned} 0 - 3(A) + 2(Ax + B) &= 4x \\ 2Ax + (2B - 3A) &= 4x. \end{aligned}$$

Match coefficients: $2A = 4 \Rightarrow A = 2$, and $2B - 3A = 0 \Rightarrow 2B = 6 \Rightarrow B = 3$. Hence $y_p = 2x + 3$ and

$$y = c_1 e^x + c_2 e^{2x} + 2x + 3$$

Solution 6 — Exponential forcing

Complementary solution. $r^2 - 1 = 0 \Rightarrow r = \pm 1$, so $y_c = c_1 e^x + c_2 e^{-x}$.

Trial. Forcing $3e^{2x}$; since $r = 2$ is not a root, $s = 0$:

$$y_p = Ae^{2x}, \quad y''_p = 4Ae^{2x}.$$

Substitute into $y'' - y$:

$$4Ae^{2x} - Ae^{2x} = 3Ae^{2x} = 3e^{2x} \Rightarrow A = 1.$$

Therefore

$$y = c_1 e^x + c_2 e^{-x} + e^{2x}$$

Solution 7 — Superposition

Complementary solution. $r^2 + 1 = 0 \Rightarrow r = \pm i$, so

$$y_c = c_1 \cos x + c_2 \sin x.$$

Split the forcing $4x + 2e^x$ into $g_1 = 4x$ and $g_2 = 2e^x$; neither overlaps y_c .

Piece 1 ($g_1 = 4x$): try $y_p^{(1)} = Ax + B$, so $(y_p^{(1)})'' = 0$ and

$$0 + (Ax + B) = 4x \Rightarrow A = 4, B = 0, \quad y_p^{(1)} = 4x.$$

Piece 2 ($g_2 = 2e^x$): try $y_p^{(2)} = Ce^x$, so $(y_p^{(2)})'' = Ce^x$ and

$$Ce^x + Ce^x = 2Ce^x = 2e^x \Rightarrow C = 1, \quad y_p^{(2)} = e^x.$$

Add the particular solutions:

$$y = c_1 \cos x + c_2 \sin x + 4x + e^x$$

Solution 8 — Exponential resonance (double root)

Complementary solution. $r^2 - 4r + 4 = (r - 2)^2 = 0 \Rightarrow r = 2$ (multiplicity 2), so

$$y_c = (c_1 + c_2 x)e^{2x}.$$

Trial. Forcing $6e^{2x}$ resonates with a *double* root, so $s = 2$:

$$y_p = Ax^2 e^{2x}.$$

Differentiate:

$$\begin{aligned}y_p' &= A e^{2x} (2x + 2x^2), \\y_p'' &= A e^{2x} (2 + 8x + 4x^2).\end{aligned}$$

Substitute into $y'' - 4y' + 4y$:

$$\begin{aligned}y_p'' - 4y_p' + 4y_p &= A e^{2x} [(4x^2 + 8x + 2) - 4(2x^2 + 2x) + 4x^2] \\&= A e^{2x} [(4 - 8 + 4)x^2 + (8 - 8)x + 2] = 2A e^{2x}.\end{aligned}$$

Set equal to the forcing: $2A e^{2x} = 6e^{2x} \Rightarrow A = 3$. Thus

$$y = (c_1 + c_2 x)e^{2x} + 3x^2 e^{2x}$$

Solution 9 — Trigonometric resonance

Complementary solution. $r^2 + 4 = 0 \Rightarrow r = \pm 2i$, so

$$y_c = c_1 \cos 2x + c_2 \sin 2x.$$

Trial. The forcing $\sin 2x$ corresponds to $\pm 2i$, a root of multiplicity 1, so $s = 1$:

$$y_p = x(A \cos 2x + B \sin 2x) = Ax \cos 2x + Bx \sin 2x.$$

Differentiating twice and simplifying gives

$$y_p'' = -4A \sin 2x + 4B \cos 2x - 4Ax \cos 2x - 4Bx \sin 2x.$$

Substitute into $y'' + 4y$; the x -terms cancel against $4y_p$:

$$y_p'' + 4y_p = -4A \sin 2x + 4B \cos 2x = \sin 2x.$$

Match: $-4A = 1 \Rightarrow A = -\frac{1}{4}$ and $4B = 0 \Rightarrow B = 0$. Hence $y_p = -\frac{1}{4}x \cos 2x$ and

$$y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4}x \cos 2x$$

Solution 10 — Resonant driven oscillator (IVP)

Complementary solution. $r^2 + 16 = 0 \Rightarrow r = \pm 4i$, so the natural response is

$$y_c = c_1 \cos 4t + c_2 \sin 4t.$$

The driving frequency (4) equals the natural frequency (4): **pure resonance**. Since $\pm 4i$ is a root of multiplicity 1, take $s = 1$:

$$y_p = t(A \cos 4t + B \sin 4t).$$

By the same differentiation pattern as Q9 (now with frequency 4),

$$y_p'' + 16 y_p = -8A \sin 4t + 8B \cos 4t = 2 \cos 4t.$$

Match: $-8A = 0 \Rightarrow A = 0$ and $8B = 2 \Rightarrow B = \frac{1}{4}$, so $y_p = \frac{1}{4}t \sin 4t$. The general solution is

$$y = c_1 \cos 4t + c_2 \sin 4t + \frac{1}{4}t \sin 4t.$$

Apply initial conditions. From $y(0) = 0$: $c_1 = 0$. Differentiate,

$$y' = 4c_2 \cos 4t + \frac{1}{4} \sin 4t + t \cos 4t \quad (\text{at } c_1 = 0),$$

and $y'(0) = 0$ gives $4c_2 = 0 \Rightarrow c_2 = 0$. Therefore

$y(t) = \frac{1}{4}t \sin 4t$

Interpretation. The envelope $\pm \frac{1}{4}t$ grows linearly in time, so the oscillations get larger without bound. This is the hallmark of pure resonance: a finite, sustained input at the natural frequency drives an unbounded response.

End of Topic 1.